AI Chatbot + CRM Platform for Real Estate Agencies

Complete phase-by-phase build prompts for Antigravity (v4)

Working name: PropBot Pro (rename freely). 14 core phases plus a new Phase 15 (A, B, C), one copy-paste prompt per phase, a master context prompt, build order, deployment blueprint, feature coverage matrix and a final QA prompt. Strict TypeScript everywhere, single Node backend, PostgreSQL + pgvector, React + Vite, deployed on Vercel + Render.

What changed in v4

- Based on a review of Kenyt, Sell.Do and Angoor. Verdict: the core stack stays; the gaps are scope and operations, not technology.
- NEW Phase 15A (platform layer): event outbox, guarded AI actions with approvals and kill switch, LLM operations (Langfuse, evals, model router), scheduled jobs, next best action. Run it right after Phase 8, before WhatsApp.
- NEW Phase 15B (growth layer, after your first pilot): no-code workflow builder, segments and campaigns, ask-your-pipeline, payments and token booking, RERA-aware copy, extra channels (email, SMS, Instagram, Messenger).
- NEW Phase 15C (last): managed voice and telephony callbacks, MCP server, enterprise security (SSO).
- Phases 1-14 updated: contacts, visitor timeline, post-visit survey, ChannelAdapter, outbox-based webhooks, materialized views, 2FA, AI controls, pilot checklist. Changes are tagged [v4] in the prompts.

The unique selling point (Phase 5 and Phase 7)

Buyer DNA + Live Intent Score + Smart Shortlist Room. The chatbot silently builds a living profile of every visitor (budget, BHK, locality, timeline, loan readiness, family size, must-haves, deal-breakers) and scores buying intent from 0 to 100 in real time. When a lead turns hot, the agent gets an instant alert with an AI-written handoff brief and next best action. The bot also creates a shareable Shortlist Room link per lead (matched properties, compare view, EMI calculator, ask-AI-about-this-property) and every click feeds back into the intent score.

Honest positioning (new in v4)

An explained intent score alone is no longer unique: Sell.Do already offers one, built from portal and call behaviour. Ours comes from the conversation itself. Position PropBot Pro as **chat-first conversion for small and mid-size developers and brokers**. Do not try to match large suites on breadth, integrations count or certifications. Kenyt's enterprise logos (large developers) will ask for security reviews and uptime promises; start with privacy controls and backups, add formal certifications only when a customer asks.

How to use this document

- 1. Open a fresh Antigravity workspace. Paste the MASTER CONTEXT prompt (Section B) first and ask it to save it as /docs/MASTER_CONTEXT.md and as the project rules file.
- 2. Run phases in the order of the build-order table (Section A2). NOTE: Phase 15A runs after Phase 8 and before Phase 9. Each prompt ends with Acceptance Criteria; do not move on until every item passes.
- 3. After each phase, ask Antigravity to update /docs/PROGRESS.md (done, pending, known issues).
- 4. If a phase is too big for one run, say: 'Complete only sub-tasks X.1 to X.3 now, then stop and report.'
- 5. Build in vertical slices: API, then its screen, then a quick end-to-end test, before moving on.
- 6. Replace anything in [SQUARE BRACKETS] with your own choices. Never paste real API keys; use .env (the prompts create .env.example only).
- 7. Start the Meta WhatsApp business verification and template approval on day one. It takes calendar time that no code can shorten.
- 8. Run each phase in Planning mode first, review the plan, then execute. Commit to git after every accepted phase.

Section A: Tech stack and architecture (v4)

Layer	Choice	Why / v4 note
Dashboard	React 18 + Vite + TypeScript (TSX) + React Router + TanStack Query + Tailwind + shadcn/ui on Vercel	Behind login so no SSR needed. v4: add React Flow for the workflow builder.
Backend (single)	Node.js 20+ + Express, strict TypeScript (ES modules, built with tsup), modular monolith (apps/api) on Render (paid, always on)	One deploy, one auth layer. No second backend needed for any competitor feature.
Database	PostgreSQL 16 + pgvector (Neon, Supabase or Render Postgres)	v4: also holds the event outbox, agent actions, approvals, audit logs, materialized views.
ORM / migrations	Drizzle ORM + drizzle-kit	Prisma is an acceptable alternative.
Queue / cache	Redis + BullMQ, workers as a separate Render Background Worker	v4: outbox dispatcher, workflow executor, repeatable nightly jobs.
Realtime	Socket.io (Redis adapter if more than one instance) + SSE for chat streaming	Live inbox, alerts, approval badge.
AI layer	apps/api/src/modules/ai with provider classes; Claude / Gemini / OpenAI via env	v4: model router with cheap/strong tiers, fallbacks and per-company cost tracking.
LLM operations (v4)	Langfuse (or similar) + own eval set run in CI	Traces, prompt versions, quality drift detection.
AI action safety (v4)	agent_actions table, permission tiers, approval queue, audit log, per-company kill switch	Anything touching price, money or promises needs a human.
Workflows (v4)	React Flow + JSON (Zod) + Node engine on BullMQ	No-code builder without a new backend.
Voice (v4, last)	Managed voice-agent/realtime provider + Exotel or similar, via VoiceProvider and TelephonyProvider	A 30-second callback needs a realtime pipeline; do not build it from scratch.
Ask your pipeline (v4)	Fixed safe query tools over Postgres under RLS	Never free-form text-to-SQL: it can leak another tenant's data.
MCP (v4, later)	TypeScript MCP SDK, read-only first	Differentiator for agencies with their own AI tools; not an MVP item.
Widget	Framework-free TypeScript compiled to one JS bundle, Shadow DOM, versioned URLs on Vercel CDN	No CSS clashes; never break old embeds.
Files	Cloudflare R2 or S3 (signed URLs)	Render disk is ephemeral; call recordings also go here.
WhatsApp	Meta WhatsApp Cloud API webhooks via ChannelAdapter	Same adapter pattern for email, SMS, Instagram, Messenger.
Auth	JWT (access + refresh) + RBAC, optional Google login, 2FA; SSO later	Roles: super_admin, company_admin, sales_manager, agent, viewer.

Layer	Choice	Why / v4 note
Monitoring	Sentry, Pino logs, uptime checks, Bull Board	v4: alerts on outbox lag, queue depth, LLM cost and errors.
Language and tooling	TypeScript strict, tsconfig project references, typescript-eslint, tsx (dev), tsup (build), Vitest	pnpm typecheck is a CI gate in every phase.
Monorepo	pnpm workspaces: apps/web, apps/api, packages/widget, packages/shared	Shared Zod schemas and inferred types for events, workflows, actions, tools.

Why TypeScript (changed from v3/v4-JS). Events, workflow nodes, agent actions, tool arguments and integrations all pass structured data between modules. Zod schemas in packages/shared are the single source of truth: z.infer gives the types, discriminated unions make the compiler flag unhandled event or node types, and the same types are used by the API, workers, dashboard and widget. The cost is a build step (tsup for the API and worker, Vite for the web app) and a typecheck gate in CI. Because the project is greenfield, no migration from JavaScript is needed.

Section A2: Build order (vertical slices)

Slice	Backend	Frontend	Phase
1	Schema, migrations, auth, roles, RLS, outbox + agent_actions tables	Login, sidebar shell, empty dashboard	1
2	Chatbot CRUD and config	8-step wizard + chatbots list	2
3	Knowledge ingestion, retrieval, model router, chat engine	Training page + embeddable widget (first demo)	3, 4
4	Lead capture, Buyer DNA, intent score	Leads page + lead drawer, contacts, visitor timeline	5, 6
5	Inventory, brochures, shortlist	Properties page + Shortlist Room	7
6	Appointments, calendar, survey	Appointments page	8
7 (NEW)	Event outbox, guarded AI actions, Langfuse, nightly jobs, next best action	Approvals page, AI Controls, event admin	15A
8	WhatsApp webhook (ChannelAdapter) + inbox	WhatsApp page	9
9	Webhooks as outbox consumers, integrations	Webhook Logs page	10
10	Analytics rollups, materialized views	Analytics page	11
11	Billing, client delivery, settings, 2FA	Users, Subscription, Settings, Client portal	12
12	Multilingual, extras, hardening, deploy	Polish, docs	13, 14
PILOT	Run with 1-2 real agencies; collect real questions into the eval set		
13 (NEW)	Workflow engine, segments, campaigns, ask-pipeline, payments, extra channels	Workflows, Campaigns, Segments	15B
14 (NEW)	Voice and telephony, MCP, SSO	Call views, AI call settings	15C

Section A3: Deployment blueprint (Vercel + Render)

Service	Platform	Notes
Dashboard (apps/web)	Vercel	Static Vite build; SPA rewrite to index.html; env VITE_API_URL
widget.js, demo/room pages	Vercel CDN	Versioned file names (widget.v1.2.0.js); long cache; never break old embeds
API (apps/api)	Render Web Service, paid, always on	Free tier sleeps and breaks chat and Meta webhooks; /healthz; graceful shutdown
Worker (same codebase)	Render Background Worker	BullMQ: outbox dispatcher, ingestion, webhooks, reminders, nightly jobs, workflow runs
PostgreSQL + pgvector	Neon, Supabase or Render Postgres	Same or nearest region to API; daily backups on a paid plan

Service	Platform	Notes
Redis	Render Key Value / Upstash	Queues, rate limits, Socket.io adapter
Files and call recordings	Cloudflare R2 or S3	Signed URLs; never store on server disk
LLM tracing (v4)	Langfuse cloud or self-hosted	Mask PII before sending traces
Region	Singapore or nearest to users	Most users are in India; keep API and DB close

Containerize the API and worker from day one (Dockerfile) so you can move to AWS, Fly.io or Railway later without a rewrite.

Section B: Master context prompt (paste this FIRST)

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ROLE
You are a senior staff engineer and product architect. You will build a production-grade, multi-tenant
SaaS: "PropBot Pro", a chat-first AI conversion platform (chatbot + CRM + lead delivery) for real estate
agencies, developers and brokers. I will give you work in phases. Always follow this master context.

POSITIONING
"Chat-first conversion for small and mid-size developers and brokers." We do NOT try to match the breadth
or certifications of large portal/CRM suites. We win on: Buyer DNA, Live Intent Score, Shortlist Room and
the hot-lead brief, all produced from the chat itself, plus fast onboarding and WhatsApp-first delivery.
[v4]

PRODUCT VISION
1. Agencies (our "clients") sign up, create chatbots, train them on their projects/inventory, and embed
them on their websites or connect them to WhatsApp.
2. The chatbot talks to property buyers 24/7, answers only from the agency's own data (RAG, no
hallucinated prices), qualifies buyers, books site visits, sends brochures and captures leads.
3. Captured leads flow into a built-in CRM and are delivered to the agency instantly (dashboard, email,
WhatsApp, webhooks) and routed to the right sales agent.
4. UNIQUE DIFFERENTIATOR: Buyer DNA + Live Intent Score (0-100) + Smart Shortlist Room + AI handoff brief
for every hot lead.

TENANCY AND ROLES
- Multi-tenant: every table carries company_id. Enforce isolation twice: (1) tenant-guard middleware sets
company_id from the JWT on every request; (2) PostgreSQL Row-Level Security policies on every tenant table
using a per-transaction setting (SET LOCAL app.company_id). Add tests proving cross-tenant reads and
writes are impossible even if application code forgets a filter.
- Roles: super_admin (platform owner), company_admin, sales_manager, agent, viewer. Permission matrix
lives in packages/shared.

TECH STACK (do not change without asking)
pnpm monorepo. Frontend: React 18 + Vite + TypeScript (TSX) + React Router + TanStack Query + Tailwind +
shadcn/ui (apps/web), deployed on Vercel. Backend: ONE Node.js 20+ + Express modular monolith in
TypeScript, ES modules (apps/api), deployed on Render. NO Python service. Database: PostgreSQL 16 +
pgvector with Drizzle ORM and drizzle-kit migrations. Redis + BullMQ (worker entrypoint as a separate
Render Background Worker). Socket.io for realtime, SSE for streamed chat answers. Zod for validation
(schemas shared from packages/shared). Widget: framework-free TypeScript compiled to a single JavaScript
bundle using Shadow DOM (packages/widget). Tests: Vitest + Supertest + Playwright. Runtime for dev/worker:
tsx; production build: tsup or esbuild to dist/ (node dist/server.js).
v4 ADDITIONS (all inside the same backend, no new language or second backend): [v4]
- Event layer: Postgres "outbox" table (domain_events) written in the same transaction as every state
change, dispatched by BullMQ workers. Webhooks, notifications, workflows, next-best-action and analytics
are consumers of this one event stream.
- Guarded AI action layer: every action an AI agent can take goes through an agent_actions table with
permission tiers, approval queue, audit log and a per-company kill switch.
- LLM operations: Langfuse (or equivalent) for traces, prompt versions and evals, plus our own eval set; a
model router (cheap model for classification/scoring, stronger model for customer answers) with fallbacks
and per-company cost tracking.
- Workflow builder: React Flow in apps/web, workflows stored as validated JSON (Zod), executed by a Node
engine on BullMQ.
- Voice: managed voice-agent / realtime speech provider + Indian telephony provider (for example Exotel)
behind a VoiceProvider interface. Never build realtime voice from scratch.
- "Ask your pipeline": fixed, safe query tools over Postgres under RLS. NO free-form text-to-SQL.
- MCP server (later): expose the CRM through the TypeScript MCP SDK with scoped tokens.

LANGUAGE RULES: Use TypeScript everywhere (apps/web, apps/api, packages/shared, packages/widget) with
"strict": true, "noUncheckedIndexedAccess": true and ES modules ("type": "module"). No `any` without a
comment explaining why; prefer `unknown` plus narrowing. Validate ALL external data (requests, webhooks,
LLM output, env vars, third-party API responses) with Zod at runtime and derive static types with z.infer
in packages/shared so the schema is the single source of truth. Event payloads, workflow JSON, agent
actions and tool arguments MUST each have a Zod schema in packages/shared, and event/action/node types are
discriminated unions so the compiler flags unhandled cases (use exhaustive `never` checks). Drizzle table
definitions provide the DB types. Do not use enums where string-literal unions work. `pnpm typecheck` (tsc
--noEmit across all workspaces, using TypeScript project references) is a required gate in CI and in every
phase. [v4]
AI code lives only in apps/api/src/modules/ai behind small provider classes (LlmProvider, Embedder,
Retriever, IntentScorer, BuyerDnaExtractor, VoiceProvider) so providers are swappable and the module could
be extracted later.

UI RULES
- Replicate the clean look of my PropBot screenshots: left sidebar (Dashboard, Chatbots, WhatsApp Bot
[New], Leads [New], Appointments [New], Analytics, Webhook Logs, Users, Subscription, Settings, Help &
Support), collapsible; top bar with global search and notifications bell; user chip showing name + role;
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white cards, soft borders, blue (#2563eb) primary, Inter font, rounded-xl cards, pill badges. v4 adds sidebar items later: Workflows, Campaigns, Approvals. **[v4]**
- Fully responsive, dark mode ready, accessible (WCAG AA), loading skeletons, empty states and error toasts everywhere.

ENGINEERING RULES
- Strict TypeScript. Zod validation on every API input. Central error handler. Pino logging. Rate limiting. Helmet. CORS allowlist per chatbot domain.
- Secrets only via environment variables; create .env.example, never real keys. Stored third-party tokens are encrypted at rest.
- Every phase must: (a) list the files it creates, (b) include tests, (c) update /docs/PROGRESS.md, (d) update /docs/API.md, (e) end by running lint + typecheck + tests and reporting results.
- Data: all schema changes through Drizzle migrations; JSONB only for flexible blobs (buyer_dna, custom fields, bot settings, event payloads); transactions for booking slots and lead-credit deductions; indexes for every filter used in the UI.
- EVENTS: every state change that matters (lead created, intent changed, visit booked, payment received...) writes a domain_event in the same transaction. Never call webhooks, notifications or workflows directly from business code; consume events instead. **[v4]**
- AI ACTIONS: AI may read freely and take reversible actions on its own, but anything involving price, discounts, payment, booking amounts, possession promises or legal claims needs human approval. No exceptions. **[v4]**
- LLM CALLS: every LLM call goes through the model router and is traced with prompt version, tokens, latency and cost per company. PII is masked before traces are stored. **[v4]**
- Seed script with demo data: 2 companies, 5 users, 3 chatbots, 200 sample leads, appointments, webhook logs.
- Privacy by design: IP, geolocation, device and page-URL tracking are opt-in per chatbot. Consent banner support, data export and delete-on-request (GDPR/CCPA/India DPDP aware).
- Marketing copy generated by AI must include the RERA registration number where required and must not make guaranteed-return or unverifiable claims. **[v4]**
- Do not invent real property prices or availability. The bot may only state facts found in the company's knowledge base or inventory; otherwise it offers a human callback.

BUILD ORDER
Vertical slices: for each feature deliver API + screen + test together. Foundations first (schema, auth, RLS), then Phase 15A (platform layer) BEFORE WhatsApp and voice, then growth features after the first real pilot. **[v4]**

WORKFLOW
Before coding each phase: write a short plan, list assumptions, then implement. After coding: run everything, fix failures, and print an Acceptance Checklist with PASS/FAIL per item. Do not start the next phase on your own.
Confirm you understood by replying with: (1) the folder structure you will create, (2) the Postgres tables list with key columns and relations (including domain_events and agent_actions), (3) the RLS plan, (4) any questions. Then wait for "Start Phase 1".

Phase 1: Foundation, multi-tenancy, auth and dashboard shell

Goal: Working monorepo, database models, login, roles, event outbox tables and the sidebar layout.

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PHASE 1 - FOUNDATION

1.1 Monorepo
- Scaffold pnpm workspace: apps/web (Vite React), apps/api (Express, with src/server.ts and src/worker.ts
entrypoints), packages/shared (Zod schemas + inferred TypeScript types, event/action/workflow unions),
packages/widget. Add ESLint, Prettier, Husky pre-commit, tsconfig.base.json (strict) with project
references per workspace, typescript-eslint, tsx for dev, tsup for builds, Dockerfile for api,
docker-compose (postgres with pgvector image, redis), .env.example, README with setup steps, /healthz
endpoint.

1.2 Database schema (Drizzle + migrations; every tenant table has company_id, created_at, updated_at;
soft-delete where noted; enable pgvector extension)
- Company (name, slug, plan, status, branding, settings)
- User (name, email, passwordHash, role, team, companyId, status: active|pending_approval|disabled,
lastLoginAt)
- Chatbot, KnowledgeSource, Conversation, Message, Lead, LeadActivity, Note, Task, Appointment,
Property/Project, Unit, Brochure, WebhookEndpoint, WebhookLog, Subscription, AuditLog, Notification
- [v4] Contact and Organization tables (a Lead links to a Contact; one Contact can have many
leads/conversations/channels).
- [v4] domain_events (id, company_id, type, payload JSONB, created_at, processed_at, attempts, last_error)
with an index on (processed_at, created_at).
- [v4] agent_actions (id, company_id, agent_name, action_type, tier, status:
proposed|approved|rejected|executed|failed, args JSONB, result JSONB, approved_by, created_at) and
company_ai_settings (kill_switch boolean, allowed_tiers, daily_action_cap).
(create full schemas, foreign keys, indexes and RLS policies now; fields will be refined later via new
migrations; JSONB for buyer_dna, custom_fields, bot settings)

1.3 Auth
- Register company + first admin, login, refresh tokens, logout, forgot/reset password (email stub),
optional Google OAuth.
- New users who join an existing company start as "pending_approval" and a company_admin approves them.
- RBAC middleware: requireRole(), requirePermission(); tenant-guard middleware that injects companyId on
every query.

1.4 Dashboard shell (apps/web)
- Collapsible sidebar: Dashboard, Chatbots, WhatsApp Bot (New), Leads (New), Appointments (New),
Analytics, Webhook Logs, Users, Subscription, Settings, Help & Support. Hide items the role cannot access.
- Top bar: global search, notification bell with unread count, avatar menu (profile, theme toggle,
logout). User chip shows name + role.
- Dashboard home: KPI cards (Total Leads, New Leads, Qualified, Converted, Active Chatbots, Today's
Visits), recent leads list, quick actions. Real API data with skeletons and empty states.

1.5 Tests
- Unit and integration tests for auth, RBAC and tenant isolation, including a test that runs a raw query
WITHOUT the app filter and proves RLS still blocks company B data.
```

ACCEPTANCE CRITERIA

- [ ] docker-compose up starts everything; migrations and seed script run
- [ ] Register, login, refresh, logout work; pending users cannot log in until approved
- [ ] Tenant isolation and RLS tests pass (including domain_events and agent_actions)
- [ ] Sidebar and top bar match the described layout and are responsive
- [ ] Lint, typecheck, tests all green

Phase 2: Chatbot builder: the 8-step wizard

Goal: Let a client create and manage chatbots exactly like your Create New Chatbot flow.

```
PHASE 2 - CHATBOT BUILDER (8-STEP WIZARD)

2.1 Chatbots list page
- Cards/table with name, type (Web / WhatsApp), status, leads count, last activity, actions (edit,
duplicate, pause, delete, embed code, open training).

2.2 Wizard at /chatbots/new with progress bar and "Step X of 8", Previous/Next buttons, per-step
validation, draft autosave, and resume.
- Step 1 Chatbot Type: radio cards "Web AI Chatbot" (Widget, Embed Code, Customizable) and "WhatsApp AI
Bot" (WhatsApp, Auto-Reply, 24/7).
- Step 2 Basic Information: Chatbot Name (required), Description.
- Step 3 Appearance: Theme cards (Default, Emerald, Purple, Rose, Amber, Custom with colour picker),
Position (Bottom Right / Bottom Left). LIVE widget preview beside the form. Also: logo upload, welcome
message, avatar, launcher text, font, border radius.
- Step 4 Behavior: Personality radio (Professional, Friendly, Technical); Response Speed (Immediate,
Natural Typing); toggles: Show Typing Indicator, Collect Email Addresses, Enable OTP Verification (verify
visitor phone via SMS/WhatsApp OTP before chat; pluggable provider, console stub).
- Step 5 Brochure and Resources: toggle "Enable Brochure Download"; upload PDF(s), per-project mapping,
"ask for phone/email before sending" option.
- Step 6 Training Data: info panel "Ready to Train Your Chatbot" and the tip card. After creation, route
to the training page (Phase 3).
- Step 7 Privacy and Tracking: IP Address Collection (off), Device Information (on), Page URL Tracking
(on), Conversion Tracking (on), Geolocation Data (off), "Privacy First Approach" info box. Consent-banner
option and data-retention days.
- Step 8 Review and Create: summary rows and button "Create and Start Training".

2.3 Extra settings after creation (tabs): General, Appearance, Behavior, Lead Capture (fields, order,
required), Business Hours and offline message, Allowed Domains, Languages, Embed Code
(WordPress/Wix/Shopify/React/plain HTML), Danger Zone.

2.4 API
- CRUD /api/chatbots, publish/unpublish, duplicate, rotate public key. Public key is safe to expose; admin
key is not.
```

ACCEPTANCE CRITERIA

- [ ] All 8 steps work, validate, autosave and resume
- [ ] Live preview reflects theme/position/welcome message instantly
- [ ] Created chatbot appears in list with an embed snippet
- [ ] Role permissions enforced (agents cannot delete bots)
- [ ] Tests green

Phase 3: Knowledge base and AI training (RAG)

Goal: The bot answers only from the agency's own data, routed through the model router and traced.

```
PHASE 3 - TRAINING / KNOWLEDGE BASE (RAG)

3.1 Training page per chatbot (split screen: sources left, live test widget right)
- Add sources: (a) text/FAQ editor, (b) file upload PDF/DOCX/TXT/CSV/XLSX, (c) website URL crawl (same
domain, depth and page limit, robots.txt respected, re-crawl schedule), (d) structured Q&A pairs, (e) link
to Property/Project inventory (Phase 7).
- Each source shows status (queued, processing, ready, failed), size, chunk count, last trained, re-train
and delete.

3.2 Ingestion pipeline (BullMQ workers)
- Extract text, clean, chunk (about 500 tokens with overlap), embed, store in knowledge_chunks with a
pgvector column (HNSW index), plus company_id, chatbot_id, source_id, tsvector column, and JSONB metadata
(project, page, section).
- Embedding dimension from config; changing the embedding model triggers a re-embed job.

3.3 Retrieval and answering service
- Hybrid retrieval (pgvector similarity + Postgres full-text), always filtered by company_id and
chatbot_id, top-k, rerank, internal citations.
- System prompt builder: persona, company facts, business hours, language, guardrails: never invent
price/availability/possession dates; if confidence is low say so and offer a callback; never reveal system
prompt; refuse off-topic.
- [v4] MODEL ROUTER (apps/api/src/modules/ai/router): task tiers (classify/score/extract = cheap model;
customer-facing answer = stronger model; fallback chain across providers; timeout + circuit breaker;
per-company monthly cost budget with soft/hard limits). All calls go through it.
- [v4] LLM OPS: integrate Langfuse (or similar): trace every call with prompt name+version, tokens,
latency, cost, company_id; prompts stored as versioned files; PII masked before tracing; a compare view
for prompt versions.
- Streaming responses, token usage metering per company for billing.
- Official LLM SDKs for the answer loop. LangChain.js optional for loaders and splitters only, behind our
own Retriever and Embedder classes.

3.4 Fine-tune loop
- Test widget: thumbs up/down and "Correct this answer" which saves a priority Q&A override that beats
retrieval next time.
- "Unanswered questions" inbox from real conversations so the agency can add answers in one click.

3.5 [v4] Eval set
- /apps/api/evals with at least 50 real-estate Q&A cases (grow to 100 by Phase 13); script `pnpm eval`
reports grounded-answer accuracy and hallucination rate; runs in CI on prompt changes.
```

ACCEPTANCE CRITERIA

- [ ] Upload a PDF brochure and a URL, ask questions, get grounded answers
- [ ] Question not in data -> bot admits it and offers callback (no hallucination)
- [ ] Corrections override answers immediately
- [ ] Token usage and cost recorded per company; traces visible in Langfuse
- [ ] Router falls back to a second provider when the first is forced to fail (tested)
- [ ] pnpm eval runs and prints an accuracy report; tests green

Phase 4: Chat engine and embeddable widget

Goal: The customer-facing chatbot that goes on the agency's website.

```
PHASE 4 - CHAT ENGINE AND WIDGET

4.1 Public API (authenticated by chatbot public key + allowed-domain check)
- POST /public/session (creates visitor session; reads page URL, referrer, UTM params, device if enabled
in privacy settings)
- POST /public/message (SSE streaming reply), GET history, POST feedback, POST /public/events (CTA
shown/clicked/started/completed/failed).
- Rate limit per session and IP; abuse and prompt-injection filter; profanity filter option. SSE with
heartbeats; disable proxy buffering and verify streaming works through Render.

4.2 Widget (packages/widget)
- Single <script src=".../widget.js" data-bot="PUBLIC_KEY"> install. Shadow DOM, under 60 KB gzipped,
lazy-loads chat UI on first open.
- Launcher bubble with proactive message; chat window with header, message list, typing indicator,
quick-reply chips, rich cards (property card with image/price/BHK/CTA), carousels, forms inside chat,
brochure download card, calendar slot picker, emoji, mobile full-screen mode, minimise, "Powered by" badge
removable on paid plans.
- Applies theme, position, personality, response speed, typing indicator, email-gate and OTP-gate options.
- Remembers returning visitors (localStorage + cookie, consent-aware).
- Proactive triggers: time on page, scroll depth, exit intent, URL rules.
- JS API: window.PropBot.open(), close(), identify({name,phone}), track(event), setContext({project}).
- [v4] Every widget interaction (page view, CTA click, property viewed, brochure click) is stored as a
visitor activity event so a per-visitor timeline can be shown later.

4.3 Conversation engine
- Intent router: faq, property_search, pricing, site_visit, brochure, human_handoff, lead_capture,
small_talk.
- Tool/function calling: search_properties, get_brochure, book_visit, capture_lead, handoff_to_agent,
calculate_emi. [v4] Tools that change data run through the guarded action layer (Phase 15A) once it
exists; until then wrap them in a stub with the same interface.
- Conversation + Message persistence with token counts, latency, sentiment per message.
- Live agent takeover: agent joins from the dashboard; bot pauses; visitor sees "A property advisor has
joined".

4.4 Test harness page /demo/:botId for quick demos.
```

ACCEPTANCE CRITERIA

- [ ] One script tag on a plain HTML page shows a working, themed chatbot
- [ ] Streaming replies, typing indicator and quick replies work on mobile and desktop
- [ ] Disallowed domain is blocked
- [ ] Agent takeover works in real time
- [ ] Tests green (API) and Playwright smoke test for widget

Phase 5: Lead capture, qualification and the UNIQUE Buyer DNA + Intent Score

Goal: Turn chats into scored, enriched leads and give agents an instant hot-lead brief.

```
PHASE 5 - LEAD CAPTURE AND BUYER DNA (UNIQUE FEATURE)

5.1 Conversational lead capture
- Configurable fields per bot (name, phone, email, budget, BHK, location, purpose, timeline, loan need).
The bot asks naturally, one question at a time, validates (phone formats incl. +91, email), and never
re-asks known info.
- Optional email gate and OTP phone verification. Mark leads "phone verified".
- Duplicate detection on phone/email (merge suggestions) and a "Check duplicates" tool.
- Capture metadata only if enabled: page URL, UTM, device, IP, geolocation. Show platform source.
- [v4] Create or link a Contact for every new lead; emit domain events lead_created / lead_updated.

5.2 BUYER DNA PROFILE (unique)
- After every message, a background extractor (cheap model via router) updates lead.buyerDna: { budgetMin,
budgetMax, bhk[], localities[], propertyType, purpose, timeline, financing, familySize, mustHaves[],
dealBreakers[], objections[], competitorsMentioned[], language, confidence per field }.
- Visual "Buyer DNA card" in the lead drawer with confidence indicators and an evolution timeline.

5.3 LIVE INTENT SCORE 0-100 (unique, explainable)
- Transparent rule + model hybrid from signals: asked price/payment plan, asked possession date, requested
site visit, shared phone, returned to chat, viewed Shortlist Room items (Phase 7), time on page, budget
fit with inventory, urgency words, number of messages, sentiment.
- Store score history. Labels: Cold (0-34), Warm (35-69), Hot (70-100). Priority maps from it. Always show
"Why this score" with top 3 reasons.
- [v4] Emit intent_changed on every material change and lead_hot when crossing the threshold. A nightly
job (Phase 15A.4) re-scores all open leads so scores decay when buyers go quiet.

5.4 HOT LEAD ALERT + AI HANDOFF BRIEF (unique)
- On lead_hot: realtime toast + notification + email/WhatsApp to the assigned agent within seconds.
- Brief: 3-line summary, Buyer DNA, objections, best-matching 3 properties, suggested opening line, and
next best action ("Call within 5 min", "Send brochure", "Offer weekend visit").
- Speed-to-lead SLA timer starts at alert.

5.5 Lead scoring config UI (Settings): editable weights and thresholds per company.
```

ACCEPTANCE CRITERIA

- [ ] A scripted demo chat produces a lead with a filled Buyer DNA and a changing intent score
- [ ] 'Why this score' explanations are shown
- [ ] Hot lead triggers alert + brief within 10 seconds
- [ ] Duplicate phone creates a merge suggestion, not a second lead
- [ ] lead_created, intent_changed and lead_hot events appear in domain_events
- [ ] Tests green

Phase 6: CRM: leads management, pipeline, tasks and follow-ups

Goal: Full lead management plus a pipeline, contacts and a visitor timeline.

```
PHASE 6 - CRM

6.1 Leads Management page (/leads)
- KPI cards: Total Leads, New Leads, Qualified, Converted (with conversion %).
- Tabs: All Leads, New, Qualified, Converted.
- Lead table with search, filters (status, priority, chatbot, time, platform), bulk select, columns: Lead Info (name, verified badge, tags), Contact, Source, Status dropdown (inline edit), Priority pill, Created, Next Follow-up, Page URL link, row actions, SLA badge. Server-side pagination, sort, saved filter views.
- Buttons: Check duplicates, Export (CSV/XLSX with chosen columns).

6.2 Lead detail drawer/page
- Tabs: Overview, Conversation transcript, Buyer DNA and Intent, Activity timeline, Notes, Tasks, Appointments, Documents.
- [v4] VISITOR TIMELINE inside the Activity tab: one chronological feed of everything this person did (pages viewed, chat messages, property views, brochure opens, Shortlist Room clicks, calls, status changes, messages sent). Built from LeadActivity + visitor events.
- Quick actions: Call, WhatsApp (wa.me prefilled), Email, Send brochure, Book site visit, Reassign, Change status, Add tag, Merge.

6.3 Pipeline (Kanban) view
- Stages: New -> Contacted -> Qualified -> Site Visit Scheduled -> Site Visit Done -> Negotiation -> Booked/Converted -> Lost (required lost reason). Drag and drop with activity logging. [v4] Every move emits lead_status_changed.

6.4 Assignment and routing
- Round-robin, weighted, by project, language, source, availability. Manual and bulk reassign. Auto-reassign if SLA breached.

6.5 Follow-ups and tasks
- Tasks with due date, type, reminders via in-app/email/WhatsApp, overdue list, "My day" view. Auto-create follow-up tasks from rules.
- Drip templates triggered by status (the full campaign builder arrives in Phase 15B).

6.6 [v4] Contacts and organizations
- /contacts list and detail: one person with multiple leads, phones, channels and conversations; optional company/organization (useful for NRI families, investors, brokers). Merge tool shared with duplicate detection.

6.7 Reporting inside CRM
- Agent leaderboard, response time, conversion by source/project, lost reasons.
```

ACCEPTANCE CRITERIA

- [ ] Table, filters, tabs, inline status change, bulk actions, export work with 10k seeded leads without lag
- [ ] Kanban drag-drop updates status, writes LeadActivity and emits an event
- [ ] Visitor timeline shows web, chat and CRM events in order for a seeded lead
- [ ] Round-robin assignment and SLA breach reassign tested
- [ ] Follow-up reminders fire via queue; tests green

Phase 7: Property inventory, brochures and the Smart Shortlist Room

Goal: Give the bot real inventory to recommend, and the unique shareable shortlist.

```
PHASE 7 - INVENTORY, BROCHURES, SHORTLIST ROOM

7.1 Inventory module (/properties)
- Projects (name, builder, RERA no., location with map pin, status, possession date, amenities, USPs,
gallery, video/3D tour link, floor plans, price sheet, payment plan, offers) and Units (tower, floor, BHK,
carpet/super area, price, availability).
- Bulk import via CSV/XLSX with validation report; optional Google Sheets sync.
- Indexed into the knowledge base automatically (Phase 3).

7.2 Property search tool for the bot
- Structured filters (budget, BHK, locality, status) + semantic ranking; returns rich property cards.

7.3 Brochure delivery
- Per-project brochure PDFs. Visitor requests it in chat; capture phone/email first if configured; send a
tracked link (chat, email or WhatsApp template). Track opens/downloads and feed the Intent Score (emit
brochure_opened).

7.4 SMART SHORTLIST ROOM (unique)
- Private, expiring link /room/:token showing 3-5 matched properties with photos, price, compare table,
map, EMI calculator, "Ask AI about this property" mini-chat, "Book site visit" and "Request callback".
- Track every interaction (views, time spent, compares, EMI use, shares) into LeadActivity, events and the
Intent Score. Agent sees which property the buyer loved most.
- Send the link automatically after qualification via chat/WhatsApp/email.

7.5 Tools: EMI calculator, price-per-sqft compare, locality insights (schools, metro, hospitals) stored
per project as editable data.
```

ACCEPTANCE CRITERIA

- [ ] Import 50 units from CSV and see them searchable by the bot
- [ ] Bot returns correct property cards for '3BHK under 1.5 Cr in [locality]'
- [ ] Brochure request is gated, delivered and tracked
- [ ] Shortlist Room link works on mobile and raises the lead's intent score on interaction
- [ ] Tests green

Phase 8: Appointments and site visits with calendar sync

Goal: Visitors book site visits in chat; agents manage them.

```
PHASE 8 - SITE VISIT APPOINTMENTS

8.1 Appointments page (/appointments)
- KPI cards: Upcoming, Today, Calendar Synced, All Appointments.
- Table: Visitor, Project (+ source), Date and time (default Asia/Kolkata), Calendar status (Synced /
Dashboard only), Status dropdown (Scheduled, Confirmed, Completed, No-show, Cancelled, Rescheduled).
Search + status filter. Buttons: Calendar Setup, Refresh.

8.2 Availability engine
- Per-agent and per-project working hours, slot length, buffer, holidays, max visits per slot, blackout
dates. Conflict detection. Round-robin among available agents.

8.3 Booking in chat
- Tool book_visit shows a slot picker card, collects missing details, confirms, and sends confirmation to
visitor and agent. Reschedule/cancel via reply or link. [v4] Emits visit_booked / visit_changed.

8.4 Google Calendar integration
- OAuth per user/company; create/update/delete events with Meet or location link; two-way status sync
where possible.

8.5 Reminders and no-show recovery
- Reminders 24h and 2h before; day-of "on my way" message with agent contact and map link; no-show
follow-up and rebooking prompt.
- Post-visit survey (rating + free text + "interested in booking?") feeding the CRM and Intent Score. [v4]
```

ACCEPTANCE CRITERIA

- [ ] Book, reschedule and cancel from chat and from dashboard
- [ ] No double-booking under concurrent requests (test it)
- [ ] Event appears in Google Calendar when connected
- [ ] Reminders sent via queue; no-show triggers follow-up; survey response updates the lead
- [ ] Tests green

Phase 9: WhatsApp AI bot and shared inbox

Goal: Automated WhatsApp conversations with human handoff. PREREQUISITE: Phase 15A (events + guarded actions) is done.

```
PHASE 9 - WHATSAPP BOT AND INBOX

Before you start: Meta business verification, a phone number and message-template approval take days or
weeks and no code speeds them up. Start that process NOW, in parallel with earlier phases. [v4]

9.1 WhatsApp Bot page (/whatsapp) with tabs: Overview, Conversations, Configuration.

9.2 Configuration tab
- Inbox automation card with toggles: Enable automation; AI auto-replies; Pause after agent reply;
Frustration handoff (sentiment detection); Auto-assign owner (round-robin). Fields: Handoff keywords
(default: agent, human, representative, complaint), Handoff message (default: "A property advisor will
continue this conversation shortly."). Button: Save automation.
- Meta API Credentials card: Permanent Access Token (encrypted, masked), Phone Number ID, Business
WhatsApp Number, Approved Details Template name, Select Chatbot dropdown, Webhook Verify Token with
Generate New and copy buttons. Button: Save Configuration.
- Webhook Configuration card: Callback URL (copy) = https://YOUR_DOMAIN/api/webhooks/meta plus the six
setup instructions (open Meta developers dashboard, add WhatsApp product, Configuration -> Edit webhook,
paste Callback URL and Verify Token, Verify and save, Manage -> subscribe to "messages").

9.3 Backend
- GET verify endpoint (hub.challenge) and POST receiver with X-Hub-Signature-256 validation, idempotency
by message id, fast 200 then queue processing.
- Handle text, interactive buttons/lists, images, documents, location, voice notes (transcribe),
delivery/read statuses. Respect the 24-hour customer-service window: outside it send only approved
templates.
- Send: text, buttons, list, image, document, template messages. Retry with backoff.
- [v4] Implement a ChannelAdapter interface (receive(), send(), normalize(), status()) and build WhatsApp
as the first adapter. Email, SMS, Instagram and Messenger (Phase 15B) reuse it.
- Create or update Lead and Contact from the WhatsApp number automatically, run the same Buyer DNA/Intent
pipeline. All outbound AI messages go through the guarded action layer: messages containing price, offers
or payment links need approval. [v4]

9.4 Shared inbox (Conversations tab)
- Three-pane inbox: list (unassigned, mine, AI-handled, needs human, closed), chat view with bot/agent
labelling, lead sidebar (Buyer DNA, intent, tasks). Take over, quick replies/templates, internal notes,
assign, snooze, close. Realtime via Socket.io. Overview tab: messages today, AI resolution rate, handoffs,
avg first response.

9.5 Opt-in/opt-out (STOP keyword), broadcast campaigns with templates to segmented leads (rate-limited,
consent-checked).
```

ACCEPTANCE CRITERIA

- [ ] Meta webhook verification succeeds with a test script
- [ ] Inbound message -> lead created -> AI replies from the linked chatbot
- [ ] 'human' keyword or frustrated tone pauses AI and alerts the team
- [ ] Agent reply pauses AI; auto-assign round-robin works
- [ ] Invalid signature rejected; duplicate message ids ignored
- [ ] An AI reply containing a price promise lands in the approval queue instead of sending
- [ ] Tests green

Phase 10: Webhooks, integrations, notifications and conversion tracking

Goal: Push leads to the client's other tools and keep a full delivery log. Webhooks are consumers of the event outbox.

```
PHASE 10 - WEBHOOKS AND INTEGRATIONS

10.1 Outbound webhooks (per company and per chatbot)
- [v4] Implemented as a consumer of domain_events (Phase 15A.1), not as direct calls from business code.
- Endpoints with URL, secret (HMAC-SHA256 signature header), custom headers, events: lead_captured,
lead_updated, lead_qualified, hot_lead, appointment_booked, appointment_changed, brochure_requested,
handoff_requested, conversation_closed, payment_received [v4].
- Payload templates with field mapping UI and "Send test event".
- Retry up to 3 attempts with exponential backoff ("Attempt 1/3"), dead-letter after failure, manual
replay.

10.2 Webhook Logs page (/webhook-logs)
- KPI cards: Total Webhooks, Success Rate, Failed, Avg Duration (whole ms).
- Filters: Lead ID, Endpoint, Status, Results per page, Start/End date, Apply, Refresh.
- Table: Timestamp, Lead Name, Chatbot, Endpoint, Event, Status badge, Duration, Attempt, Actions
(request/response drawer with redacted secrets, replay).

10.3 Native integrations (adapter pattern, each optional). Order of priority: webhooks, Google Sheets, ONE
popular CRM (HubSpot or Zoho), then the rest on client demand. [v4]
- CRM sync: Zoho CRM, HubSpot, Salesforce, LeadSquared, Google Sheets, Zapier/Make, generic REST.
- Messaging: email (SMTP/SendGrid/SES), SMS (Twilio/MSG91), WhatsApp templates.
- Ads/analytics: GA4, Google Ads, Meta Pixel/Conversions API events from the widget when conversion
tracking is enabled, with consent mode.
- Lead ingestion: Facebook/Instagram Lead Ads, Google Ads lead forms, portals (99acres, MagicBricks,
Housing) via webhook/email-parse endpoint, public REST API with API keys and docs. New leads trigger the
instant WhatsApp "welcome + qualify" message.

10.4 Notification center
- In-app bell, email digest, WhatsApp/SMS alerts, user preferences, quiet hours. Fed by domain events.
[v4]
```

ACCEPTANCE CRITERIA

- [ ] Lead capture triggers a signed webhook via the outbox; failure retries 3 times then dead-letters
- [ ] Logs page filters and replay work
- [ ] Google Sheets (or generic REST) sync works with a mock server in tests
- [ ] Conversion pixel events fire only when consent is given
- [ ] Tests green

Phase 11: Analytics, attribution and forecasting

Goal: Dashboards that prove ROI to the agency.

```
PHASE 11 - ANALYTICS

11.1 Analytics Dashboard (/analytics)
- Header buttons: Forecast, Attribution, date-range picker, Export Report (PDF/CSV).
- KPI cards with real trend deltas (vs previous period; no hard-coded numbers): Total Conversations, Leads Generated, Conversion Rate, Avg Response Time, Satisfaction Rate, Active Chatbots.
- CTA funnel: Shown -> Clicked -> Started -> Completed -> Failed.
- Tabs: Performance, Engagement (messages per chat, drop-off, peak hours heatmap), Top Queries (clustered questions, unanswered list), Trends.

11.2 Real-estate-specific insights
- Funnel: Visitor -> Chat -> Lead -> Qualified -> Visit booked -> Visit done -> Booked.
- Lead quality by source/project/campaign, cost per lead (manual spend input), response time vs conversion, hot-lead speed-to-lead, most asked objections, demand by locality/BHK/budget band.
- [v4] AI cost and quality panel: LLM cost per lead, approval-queue stats, eval accuracy trend.

11.3 Attribution: first-touch, last-touch, linear using UTM and page URL; campaign/ad performance table.

11.4 Forecast: explainable 30-day forecast (moving average + seasonality) with confidence band, labelled as an estimate.

11.5 Scheduled reports: weekly/monthly email per company; super_admin cross-company analytics.

11.6 Performance
- [v4] Use Postgres materialized views and/or daily rollup tables, refreshed by BullMQ repeatable jobs (Phase 15A.4). Dashboard must load under 1.5 s with 100k conversations.
```

ACCEPTANCE CRITERIA

- [ ] All numbers come from real data and match raw counts in tests
- [ ] Date range, chatbot filter and export work
- [ ] Empty state shown (not fake numbers) for new accounts
- [ ] Tests green

Phase 12: Admin: users, teams, subscription and settings

Goal: Company management, billing and lead delivery controls for the platform owner.

```
PHASE 12 - USERS, SUBSCRIPTION, SETTINGS, LEAD DELIVERY

12.1 User Management (/users)
- KPI cards: Total Users, Active Users, Pending Approval, Companies.
- Table: avatar, name, email, Company, Role badge, Team, Status, Joined, Actions. Manage drawer: role,
team, chatbots/projects access, working hours, lead capacity, 2FA reset. Invite by email. Approve/reject
pending users. super_admin sees all companies; company_admin only theirs.
- Audit log of sensitive actions (login, role change, export, delete, AI approvals [v4]).

12.2 LEAD DELIVERY TO CLIENTS (core business requirement)
- Agency/Reseller mode: manage multiple client companies, assign each chatbot to a client, deliver
captured leads by dashboard + email + WhatsApp + webhook, with per-client delivery rules (working hours,
round-robin members, duplicate window).
- Optional lead-credit model: lead balance per client; each delivered qualified lead consumes a credit
(transaction); low-balance alerts; invoice-ready usage report.
- Client portal: restricted view where the client sees only their leads, appointments and analytics.

12.3 Subscription (/subscription)
- Plans (Free/Trial, Starter, Growth, Enterprise) with limits: chatbots, conversations/month, AI tokens,
users, WhatsApp numbers, storage, feature flags (white-label, API access, voice, workflows [v4]). Usage
meters, upgrade/downgrade, invoices. Razorpay and Stripe via a payment-provider interface (stub in dev).
Enforce limits in API middleware with upgrade prompts.

12.4 Settings (/settings)
- Company profile, branding and white-label, business hours, lead scoring weights, pipeline stages, tags,
custom lead fields, notification defaults, API keys, data retention and export/delete tools.
- Security: 2FA (TOTP) for all roles, session management, IP allowlist for enterprise. [v4] SSO
(SAML/OIDC) is deferred to Phase 15C.
- [v4] AI Controls: kill switch, allowed action tiers, daily action cap, approval routing (who approves
what).
```

ACCEPTANCE CRITERIA

- [ ] Pending users can be approved/rejected; roles enforced everywhere
- [ ] A lead from a chatbot assigned to Client A is visible only to Client A, delivered through their channels
- [ ] Plan limits block over-quota actions with an upgrade message
- [ ] Client portal shows only that client's data (tested)
- [ ] Tests green

Phase 13: Multilingual and advanced AI extras

Goal: Competitive AI behaviours. Voice calling moved to Phase 15C; WhatsApp voice-note transcription stays here.

```
PHASE 13 - ADVANCED AI EXTRAS

13.1 Multilingual
- Auto-detect visitor language and reply in it (English, Hindi, Hinglish, Marathi, Tamil, Telugu, Kannada,
Bengali, Gujarati, Arabic and others). Per-bot allowed languages; translatable UI strings; agent sees
translated transcript.

13.2 Voice notes (text-to-speech/voice mode in the widget is optional)
- Speech-to-text for WhatsApp voice notes, provider-pluggable. [v4] Outbound AI calling is NOT built here;
see Phase 15C.

13.3 Smart behaviours (all outbound actions go through the guarded action layer)
- Proactive nudges (last units, offer ending), cross-sell similar projects, objection-handling playbooks
editable by the agency, competitor comparison using only agency-provided data.
- Sentiment and frustration detection driving handoff. Auto-summaries and tags saved on the lead.
- Chat-to-visit nudges: if intent is high and no visit booked, offer top 3 slots.
- Re-engagement of dormant leads on day 1/3/7/30 with personalised hooks from Buyer DNA (templates,
consent-checked).

13.4 Agent copilot (dashboard)
- "Suggest reply", "Summarise this lead", call-script generator from Buyer DNA, objection rebuttals.

13.5 Safety and quality
- Prompt-injection defences, PII masking in logs, hallucination guard, content moderation, human review
queue for low-confidence answers, eval set of 100 real-estate Q&A with automated accuracy report in CI.
```

ACCEPTANCE CRITERIA

- [ ] Hindi/Hinglish chat works end to end and creates a lead
- [ ] Voice note on WhatsApp is transcribed and answered
- [ ] Re-engagement sequence runs on a schedule and respects opt-out
- [ ] Eval script reports grounded-answer accuracy on 100 cases
- [ ] Tests green

Phase 14: Hardening, QA, deployment and documentation

Goal: Make it secure, fast, tested, deployed and documented, ready for the first real pilot.

```
PHASE 14 - PRODUCTION READINESS

14.1 Security
- OWASP Top 10 review; input validation everywhere; secure cookies; CSRF where applicable; encrypted
secrets at rest; signed URLs for files; per-tenant rate limits; audit logs; dependency scan; security
headers; widget XSS-safe rendering of AI output.
- [v4] Specific tests: AI action layer cannot be bypassed (try tool calls with the kill switch on);
approval-tier actions never execute without an approver; trace storage contains no raw phone numbers or
emails.

14.2 Compliance
- Consent banner, privacy policy generator text for agencies, data export + delete flows, retention jobs,
DPA-ready docs. Start with privacy controls and backups; formal certifications (ISO 27001, SOC 2) only
when customers ask. [v4]

14.3 Performance and reliability
- EXPLAIN ANALYZE on key queries; connection pooling; pagination everywhere; cached public widget config;
Bull Board (admin only); health checks, graceful shutdown, structured logs, Sentry, uptime checks.
- [v4] Alerts on: outbox lag, queue depth, webhook failure rate, LLM error rate and cost spikes.
- Load test the public chat API (target 200 concurrent chats) and report results.

14.4 Testing
- Unit, integration, and Playwright E2E: signup -> create bot -> train -> embed -> chat -> lead appears ->
assign -> book visit -> webhook delivered -> analytics updated. Coverage report.

14.5 DevOps
- Dockerfile for api and worker (multi-stage: build with tsup, run compiled dist/ on Node 20),
docker-compose prod-like, GitHub Actions CI (lint, typecheck, test, eval, build, migration check), staging
and prod. Deploy guide: apps/web on Vercel (SPA rewrites, VITE_API_URL), apps/api as paid always-on Render
Web Service, worker as Render Background Worker, Postgres with pgvector, Redis, R2/S3. render.yaml,
zero-downtime migrations, backup and restore drill, versioned widget.js on CDN. The Render free tier
sleeps and must not be used in production.

14.6 Documentation
- README, ARCHITECTURE.md, API.md (OpenAPI at /api/docs), WIDGET_INSTALL.md, WHATSAPP_SETUP.md,
ADMIN_GUIDE.md, CLIENT_GUIDE.md, RUNBOOK.md, CHANGELOG.md, [v4] EVENTS.md (event catalog) and
AI_ACTIONS.md (tiers and approval rules). In-app Help & Support page.

14.7 Final demo
- Seed a polished demo company with realistic Indian real estate projects, 200 leads, visits, analytics
history, and a script that records the golden-path demo.

14.8 [v4] Pilot checklist
- Onboard 1-2 real agencies. Collect real questions into the eval set, track wrong-answer reports,
response times and approval-queue volume before starting Phase 15B.
```

ACCEPTANCE CRITERIA

- [ ] Full E2E suite green; coverage report generated
- [ ] No high/critical dependency vulnerabilities
- [ ] Load test report attached
- [ ] One-command deploy guide verified
- [ ] All docs present; pilot checklist ready

Phase 15A: Platform layer (RUN AFTER PHASE 8, BEFORE PHASE 9)

Goal: Event backbone, guarded AI actions, LLM operations and scheduled jobs. Everything later depends on this.

```
PHASE 15A - PLATFORM LAYER

15.1 Event outbox
- domain_events written inside the same DB transaction as the state change. A dispatcher worker claims
unprocessed rows with SELECT ... FOR UPDATE SKIP LOCKED, fans out to BullMQ queues per consumer, marks
processed, retries with backoff, parks failures as dead events.
- Event catalog in packages/shared with a Zod schema per type: lead_created, lead_updated, intent_changed,
lead_hot, lead_status_changed, message_received, visit_booked, visit_changed, brochure_opened,
room_viewed, handoff_requested, payment_received, survey_completed.
- Consumers: notifications, webhooks, next-best-action, analytics rollups; workflows later (15B). Each
consumer is idempotent (dedupe by event id).
- Admin tool: view, filter and replay events. Document the catalog in /docs/EVENTS.md.
- Refactor Phases 4-8 to emit events instead of calling consumers directly.

15.2 Guarded AI action layer
- All AI tool calls and AI-initiated side effects go through executeAgentAction(companyId, agent,
actionType, args).
- Permission tiers: T0 read-only (search, summarise): auto. T1 reversible and low risk (add tag, create
task, draft a message, update Buyer DNA): auto. T2 sensitive (send a message containing price, discount,
offer, payment link or possession promise; mark Booked; change assignment of a hot lead): needs human
approval. T3 never by AI (refunds, contract terms, deleting data).
- Approval queue UI (/approvals): see proposed action, context, who must approve, approve/edit/reject;
expiry for stale items. Realtime badge in the sidebar.
- Per-company kill switch (instant), allowed tiers, daily caps, per-lead rate caps. Agents run under a
service identity subject to the same RBAC and RLS as a user.
- Full audit trail in agent_actions + AuditLog (who/what/why, args, result, approver).

15.3 LLM operations
- Langfuse (cloud or self-hosted): traces, prompt versions, eval runs. Router and eval script from Phase 3
wired in. Cost dashboard per company. CI fails if eval accuracy drops more than a set threshold.

15.4 Scheduled jobs (BullMQ repeatable)
- Nightly intent re-scoring with decay; daily rollups and materialized view refresh; reminder sweeps;
retention and cleanup; stale-approval expiry; Meta/Google token refresh.

15.5 Next best action
- Consumer of intent_changed, visit_booked, room_viewed, message_received that writes a recommended action
on the lead ("Call within 5 min", "Send brochure", "Offer weekend visit") and shows it in the lead drawer
and hot-lead brief. Executing it goes through 15.2.
```

ACCEPTANCE CRITERIA

- [ ] Events are written transactionally and processed exactly once per consumer (tested with forced crashes)
- [ ] A T2 action never executes without approval; kill switch blocks T0-T2 instantly
- [ ] Every LLM call has a trace with cost; eval regression fails CI
- [ ] Nightly job re-scores leads and decays idle ones
- [ ] Next best action appears on a hot lead
- [ ] Tests green

Phase 15B: Growth layer (RUN AFTER THE FIRST PILOT)

Goal: Workflow builder, segments and campaigns, payments, extra channels and smarter reporting.

```
PHASE 15B - GROWTH LAYER

15.6 Workflow builder (/workflows)
- React Flow canvas. Workflows saved as JSON validated by a Zod schema; versioned; draft/active.
- Nodes: Trigger (any domain event), Condition (intent, status, source, Buyer DNA fields), Action (send
WhatsApp template/email/SMS, assign, create task, add tag, change status, call webhook, wait N hours),
Branch, End.
- Executor: Node engine on BullMQ with delayed jobs; one run record per event with step log; dry-run/test
mode; pause and cancel; loop guards. Every action calls the guarded action layer, so approval tiers still
apply.
- "Describe in plain English": the AI drafts workflow JSON from a sentence; the user must review and
activate it. AI never activates workflows by itself.
- Templates: "Welcome + qualify new lead", "Hot lead alert", "No reply in 24h", "Post-visit survey",
"Dormant lead re-engagement".

15.7 Segments and campaigns
- Saved segments (filters on intent label, Buyer DNA fields, status, source, project, last activity) with
live counts.
- Campaign builder: drip sequences across WhatsApp/email/SMS, schedule, frequency caps, consent and
opt-out checks, send windows, simple A/B on message variants, per-campaign stats.

15.8 Ask your pipeline
- A chat box in the dashboard answering questions such as "which hot leads have not been called today?".
The LLM may only call fixed query tools (list_leads, lead_stats, visits_summary, source_performance) that
run parameterised SQL under RLS. NO model-written SQL. Answers cite the returned records (click to open
them).

15.9 Payments and token booking
- Razorpay (and Stripe) payment links for booking amounts on a lead/unit; payment status webhooks;
receipts; refunds handled manually by admins. Emit payment_received; stage moves to Booked only after
confirmed payment or admin action. Creating or sending a payment link is a T2 action (approval).

15.10 RERA-aware marketing copy
- AI copy generator for ads/WhatsApp broadcasts that inserts the project RERA number, blocks
guaranteed-return or unverifiable claims, and flags content for review. Copy sources come only from
project data.

15.11 Extra channels (use the ChannelAdapter from Phase 9)
- Order: email (inbound + outbound), SMS, Instagram DMs, Facebook Messenger. Telegram only on request. All
feed the same Contact, Conversation and Intent pipeline.
```

ACCEPTANCE CRITERIA

- [ ] A workflow built on the canvas runs on a live event, logs each step and respects approval tiers
- [ ] Segment counts match raw SQL in tests; a campaign skips opted-out contacts
- [ ] Ask-your-pipeline returns cited answers and cannot read another tenant's data (tested with adversarial prompts)
- [ ] Payment link flow works in test mode and emits payment_received
- [ ] Generated ad copy always includes the RERA number and blocks guaranteed-return wording
- [ ] Email and SMS channels create leads through the same pipeline; tests green

Phase 15C: Voice, MCP and enterprise (LAST)

Goal: Most operationally heavy items. Only start when clients are asking for them.

```
PHASE 15C - VOICE, MCP, ENTERPRISE

15.12 Voice calling (feature flag, per company)
- Use a managed voice-agent or realtime speech provider behind VoiceProvider plus an Indian telephony
provider (for example Exotel) behind TelephonyProvider. Do NOT build realtime speech from scratch.
- Instant callback: on lead_created from an ad or portal, if the lead gave consent, call within about 30
seconds (inside permitted calling hours; honour DND/telecom rules). The AI states at the start that it is
an AI assistant. Offer to transfer to a human or book a visit.
- Webhooks from the provider into Node; call records in Postgres; recordings in R2; transcription and call
summary as background jobs; call transcript feeds Buyer DNA and the Intent Score; manual call logging for
human agents too.
- Safety: consent log, opt-out phrase handling, call caps, kill switch applies. Discussion of price beyond
published data escalates to a human.
- Click-to-call and call tracking numbers per campaign (optional).

15.13 MCP server
- Expose the CRM through the TypeScript MCP SDK. Start READ-ONLY (leads, pipeline stats, visits).
Per-company scoped tokens, rate limits and audit logs. Any write tool goes through the guarded action
layer.

15.14 Enterprise security
- SSO (SAML/OIDC), enforced 2FA policy, audit-log export, secrets rotation, IP allowlist, SLA/uptime page,
security questionnaire answers pack. Pursue a formal certification only when a customer requires it.

15.15 Optional extras (skip unless asked): service-desk tickets, video call/screen share, non-real-estate
verticals.
```

ACCEPTANCE CRITERIA

- [ ] Test call: callback triggers within the target time for a consenting test lead, with AI disclosure
- [ ] Transcript and summary appear on the lead and change the intent score
- [ ] MCP read-only tools work from an external client and cannot see other tenants
- [ ] SSO login works for a test identity provider
- [ ] Tests green

Section C: Feature coverage matrix (v4)

Every screen from your PDF, every common Kenyt-style capability, and the gaps found in Kenyt, Sell.Do and Angoor, mapped to the phase that builds it.

Feature	Source	Phase
Create New Chatbot wizard, themes, preview, personality, typing indicator	Your PDF	2, 4
Email collection, OTP verification	Your PDF	2, 5
Brochure download, training data, privacy toggles	Your PDF	2, 3, 7, 14
WhatsApp automation, handoff, Meta credentials	Your PDF	9
Leads table, KPIs, filters, SLA, duplicates, export	Your PDF	5, 6
Site-visit appointments + Google Calendar, post-visit survey	Your PDF / Kenyt	8
Analytics, CTA funnel, attribution, forecast	Your PDF	11
Webhook logs with retries	Your PDF	10
User management, approvals, subscription, settings	Your PDF	1, 12
24/7 grounded AI chatbot, lead qualification, property search	Kenyt	3, 4, 5, 7
Built-in CRM, pipeline, tasks, routing; contacts and organizations	Kenyt / Your request	6
Visitor activity timeline	Kenyt	4, 6
Lead delivery to clients, client portal, lead credits	Your request	12
Buyer DNA, Intent Score, hot-lead brief, Shortlist Room	Unique	5, 7
Overnight re-scoring and next best action	Sell.Do	15A
Event backbone feeding webhooks, alerts, workflows	Sell.Do / Angoor	15A
AI acts within limits: tiers, approvals, audit, kill switch	Sell.Do	15A
Prompt versions, LLM logs, evals, multi-provider routing	Angoor	3, 15A
Multilingual (Hindi, Hinglish and more), voice notes	Kenyt	13
Drip campaigns and segments by intent	Kenyt	15B
No-code workflow builder (and plain-English drafting)	Kenyt / Angoor	15B
Ask questions of your pipeline with citations	Sell.Do	15B
Payments and token booking (Razorpay, Stripe)	Kenyt	15B
RERA-aware marketing copy	Sell.Do	15B
Email, SMS, Instagram, Messenger channels	Kenyt / Angoor	15B
Voice callback in about 30 seconds, call transcription and scoring	Sell.Do / Kenyt	15C
MCP server	Sell.Do	15C
SSO, enterprise security (certifications only on demand)	Sell.Do	15C
Service desk, video call, other industries	Kenyt	Skipped on purpose

Section D: Final verification prompt (run after Phase 14, again after 15A, 15B, 15C)

```
FINAL AUDIT
Act as an independent QA lead. Read /docs/MASTER_CONTEXT.md, /docs/PROGRESS.md and the entire codebase.
1. Build a table of every feature promised in all phase prompts (1-14 and 15A/B/C that are done so far)
and mark each: Implemented / Partial / Missing, with file paths as evidence.
2. Run the full test suite, lint, typecheck, pnpm eval and the Playwright golden-path. Report failures.
3. Try to break tenant isolation, webhook signature validation, rate limits and prompt injection. [v4]
Also try to: bypass the AI action layer, get a T2 action to run without approval, read another tenant's
data through ask-your-pipeline, and leak PII into LLM traces. Report findings.
4. Check the UI against the sidebar/screens spec and list visual or UX gaps.
5. Produce /docs/GAP_REPORT.md with a prioritised fix list, then fix all Missing and Partial items, re-run
everything, and print the final scorecard.
```

Tips for best results in Antigravity

- If the agent gets stuck, paste only the failing error plus the relevant sub-task number, not the whole phase.
- Start every new session with: 'Read MASTER_CONTEXT.md and PROGRESS.md, then continue with Phase N.'
- Keep the eval set growing from real pilot questions; it is your best defence against wrong prices reaching buyers.
- Use real data for the demo: 3 real-looking projects, a brochure PDF, and your own WhatsApp test number in the Meta sandbox.